

INDIAN STATISTICAL INSTITUTE, BANGALORE CENTRE
B.MATH - First Year, First Semester, 2010-11
Probability Theory-I, Backpaper Examination
Answer any 5 questions. All questions carry equal marks.

1. Suppose there are 3 chests each having 2 drawers. The first chest has a gold coin in each drawer, the second chest has a gold coin in one drawer and a silver coin in the other drawer, and the third chest has a silver coin in each drawer. A chest is chosen at random and a drawer is opened. If that drawer contains a gold coin what is the probability that the second drawer has a silver coin?
2. A student is taking a multiple choice exam in which each question has 4 possible answers, exactly one of which is correct. If the student knows the answer she selects the correct answer. Otherwise she selects one answer at random from the 4 possible answers. (There is no negative marking.) Suppose that there are 100 questions carrying 1 mark each in the exam and the student knows the answer to 70% of the questions. Let X denote the total marks that the student will get in this exam. What is the probability distribution of X ?
3. Let X and Y be jointly distributed discrete random variables and let g and h be functions such that $P(X = x, Y = y) = g(x)h(y)$ for all x and y .
 - (a) Express $f_X(x)$ in terms of g and h .
 - (b) Express $f_Y(y)$ in terms of g and h .
 - (c) Show that X and Y are independent.
4. There are three coins in a box. These coins, when tossed, will come up heads with respective probabilities of 0.3, 0.5 and 0.7. A coin is randomly picked from the box and tossed 10 times independently. Let X denote the number of heads obtained in these 10 tosses. Find the probability distribution of X .
5. Suppose $X \sim \text{Poisson}(\lambda)$ and $Y|X = k$ has the discrete uniform distribution on the set of integers $-k \leq y \leq k$.
 - (a) Show that $|Y| \leq X$ and hence $E(Y^2) < \infty$.
 - (b) Find $E(Y)$ and $Var(Y)$.

6. Let X be a continuous random variable with p.d.f.

$$f(x) = \begin{cases} c_1 x & \text{for } 0 \leq x < 1/3; \\ c_2(1-x) & \text{for } 1/3 \leq x < 1; \\ 0 & \text{otherwise,} \end{cases}$$

where c_1 and c_2 are constants which make f a continuous function.

- (a) Find c_1 and c_2 .
- (b) Find the c.d.f. of X .
- (c) Find $P(X \geq 2/3)$ and $P(1/3 < X \leq 2/3)$.

7. Let X be a continuous random variable having the p.d.f

$$f(x) = \begin{cases} \lambda \exp(-\lambda x) & \text{if } x > 0; \\ 0 & \text{otherwise.} \end{cases}$$

Find the p.d.f. of $Y = X^{1/\alpha}$ where $\alpha > 0$ is a constant.

8. Let p_n , $n = 0, 1, 2, \dots$ be non-negative numbers such that $\sum_{n=0}^{\infty} p_n = 1$. Suppose $U \sim U(0, 1)$. Define $X = 0$ if $U \leq p_0$, $X = 1$ if $p_0 < U \leq p_0 + p_1$, and in general $X = n$ if $\sum_{i=0}^{n-1} p_i < U \leq \sum_{i=0}^n p_i$ for $n \geq 2$. Find the probability distribution of X .

9. Let $X \sim \text{Binomial}(100, 0.1)$. Using the normal approximation

- (a) find $P(X = 10)$, and
- (b) find $P(3 < X \leq 16)$.

10. A balanced die is rolled n times independently where $n \geq 2$. Let X denote the number of times 6 dots show up and Y denote the number of times 5 dots show up in these n rolls.

- (a) What is the joint probability distribution of (X, Y) ?
- (b) Find the probability distribution of $Z = X + Y$.
- (c) Find $E(Z)$, $Var(Z)$ and $Cov(X, Z)$.

Cumulative Normal Probability Tables (Z-Values)

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55962	0.56356	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71226	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91308	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449
1.7	0.95543	0.95637	0.95728	0.95818	0.95907	0.95994	0.96080	0.96164	0.96246	0.96327
1.8	0.96407	0.96485	0.96562	0.96638	0.96712	0.96784	0.96856	0.96926	0.96995	0.97062
1.9	0.97128	0.97193	0.97257	0.97320	0.97381	0.97441	0.97500	0.97558	0.97615	0.97670
2.0	0.97725	0.97778	0.97831	0.97882	0.97932	0.97982	0.98030	0.98077	0.98124	0.98169
2.1	0.98214	0.98257	0.98300	0.98341	0.98382	0.98422	0.98461	0.98500	0.98537	0.98574
2.2	0.98610	0.98645	0.98679	0.98713	0.98745	0.98778	0.98809	0.98840	0.98870	0.98899
2.3	0.98928	0.98956	0.98983	0.99010	0.99036	0.99061	0.99086	0.99111	0.99134	0.99158
2.4	0.99180	0.99202	0.99224	0.99245	0.99266	0.99286	0.99305	0.99324	0.99343	0.99361
2.5	0.99379	0.99396	0.99413	0.99430	0.99446	0.99461	0.99477	0.99492	0.99506	0.99520
2.6	0.99534	0.99547	0.99560	0.99573	0.99585	0.99598	0.99609	0.99621	0.99632	0.99643
2.7	0.99653	0.99664	0.99674	0.99683	0.99693	0.99702	0.99711	0.99720	0.99728	0.99736
2.8	0.99744	0.99752	0.99760	0.99767	0.99774	0.99781	0.99788	0.99795	0.99801	0.99807
2.9	0.99813	0.99819	0.99825	0.99831	0.99836	0.99841	0.99846	0.99851	0.99856	0.99861
3.0	0.99865	0.99869	0.99874	0.99878	0.99882	0.99886	0.99889	0.99893	0.99896	0.99900
3.1	0.99903	0.99906	0.99910	0.99913	0.99916	0.99918	0.99921	0.99924	0.99926	0.99929
3.2	0.99931	0.99934	0.99936	0.99938	0.99940	0.99942	0.99944	0.99946	0.99948	0.99950
3.3	0.99952	0.99953	0.99955	0.99957	0.99958	0.99960	0.99961	0.99962	0.99964	0.99965
3.4	0.99966	0.99968	0.99969	0.99970	0.99971	0.99972	0.99973	0.99974	0.99975	0.99976
3.5	0.99977	0.99978	0.99978	0.99979	0.99980	0.99981	0.99981	0.99982	0.99983	0.99983
3.6	0.99984	0.99985	0.99985	0.99986	0.99986	0.99987	0.99987	0.99988	0.99988	0.99989
3.7	0.99989	0.99990	0.99990	0.99990	0.99991	0.99991	0.99992	0.99992	0.99992	0.99992
3.8	0.99993	0.99993	0.99993	0.99994	0.99994	0.99994	0.99994	0.99995	0.99995	0.99995
3.9	0.99995	0.99995	0.99996	0.99996	0.99996	0.99996	0.99996	0.99996	0.99997	0.99997
4.0	0.99997	0.99997	0.99997	0.99997	0.99997	0.99997	0.99998	0.99998	0.99998	0.99998